



AMERICAN INTERNATIONAL UNIVERSITY-BANGLADESH

Faculty of Science and Technology
Department of Computer Science
CSC 3215: Web Technologies

Semester: Summer 2025-26

Group No:9

Section:B

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Project Title

Hospital Bed Management and Patient Admission System

Short Description

The Hospital Bed Management System is a database driven platform designed to streamline bed allocation, patient admissions, and hospital operations. By linking core entities including Hospitals, Doctors, Beds, Patients, Admissions, Payments, Notifications, and Feedback the system digitizes the operational lifecycle. Hospitals manage bed inventories, prices, and attending doctors, while patients request admissions, track real-time bed availability, and complete payment transactions directly through the platform. Automated notifications alert users of status changes, and integrated feedback mechanisms collect post treatment patient experiences. Overall, the platform eliminates manual record-keeping, minimizes allocation delays, improves resource transparency, and elevates the overall standard of patient care.

Introduction

Efficient resource allocation in healthcare facilities is essential for providing timely medical interventions and ensuring operational efficacy. The Hospital Bed Management System provides a centralized digital infrastructure designed to optimize hospital workflow and automate patient care logistics. By organizing relational data across key entities such as Hospitals, Beds, Doctors, Patients, and Admissions, the system simplifies the process of assigning available beds based on patient needs, medical specialties, and pricing categories.

Patients can search for hospitals, request admissions, receive automated system notifications, and process secure payments seamlessly. Concurrently, medical staff can oversee doctor schedules, monitor real-time bed statuses, and handle admissions without administrative overhead. Additionally, the system incorporates a feedback framework to capture patient reviews, helping healthcare providers continuously assess service quality. Ultimately, this system mitigates manual errors, reduces wait times, and provides a scalable framework to modernize hospital administration.

Background Study

Traditional hospital operations often rely heavily on manual bed tracking and fragmented communication systems, leading to delayed patient admissions, misallocated resources, and administrative bottlenecks. During periods of high patient volume, non-centralized data models fail to provide real-time updates regarding bed availability, pricing, or physician assignments, which negatively impacts critical care delivery. Modern healthcare facilities require structured, database-driven software architectures to handle concurrent admissions and payment tracking efficiently. By establishing relational structures between patients, doctors, beds, and financial records, digital systems eliminate manual record redundancies. Developing an automated, interconnected Bed Management System directly addresses these logistical challenges, ensuring transparent data management, faster response times, and improved overall healthcare operational efficiency.

ER Diagram

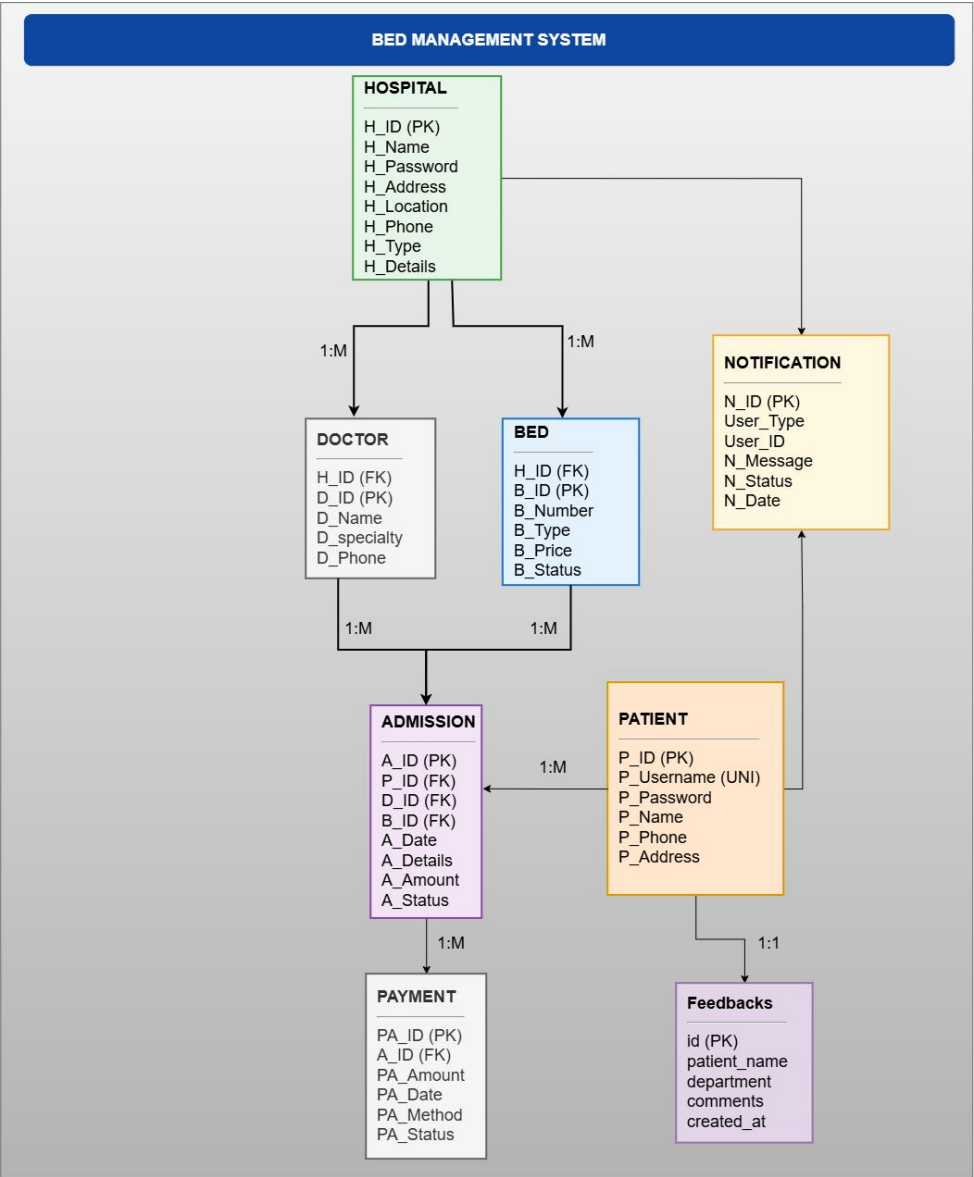
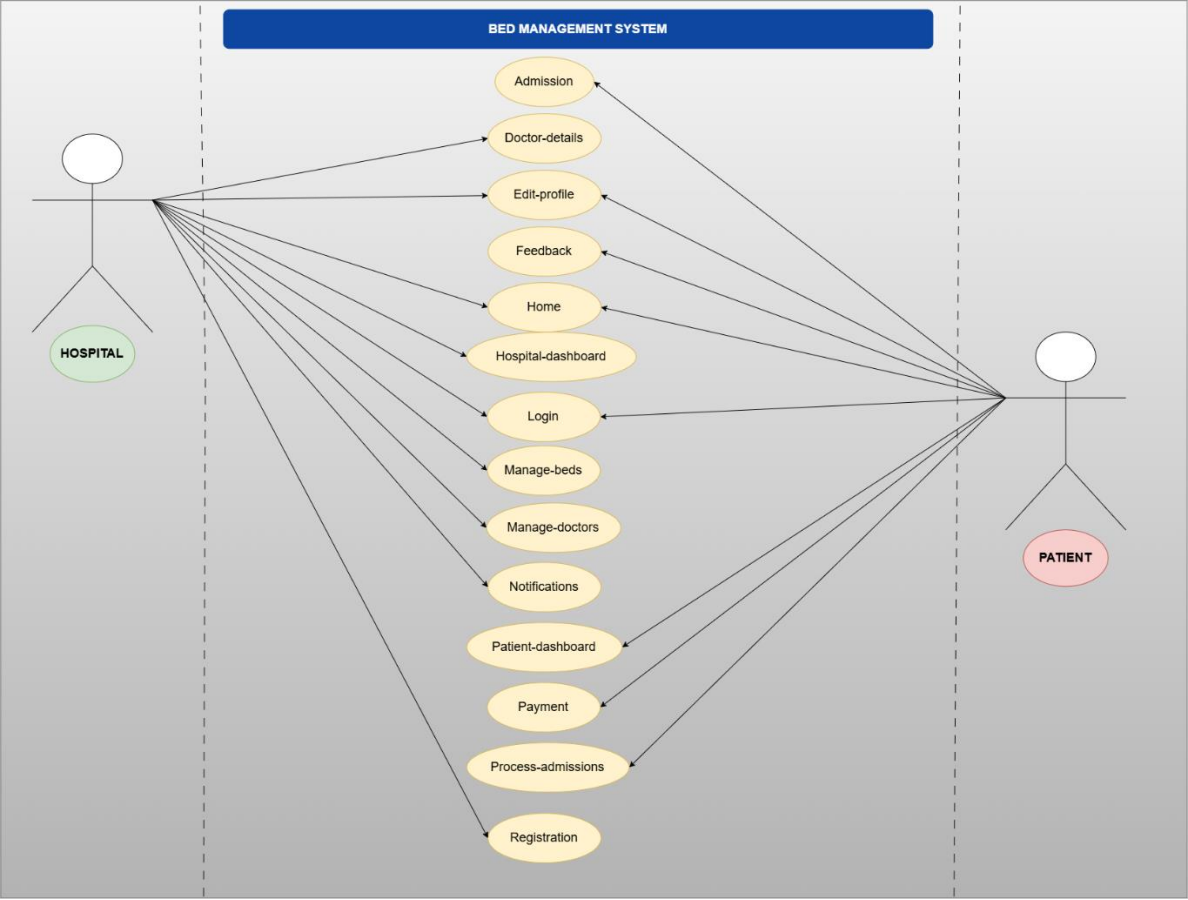


Figure: Relational Database Schema for Bed Management System

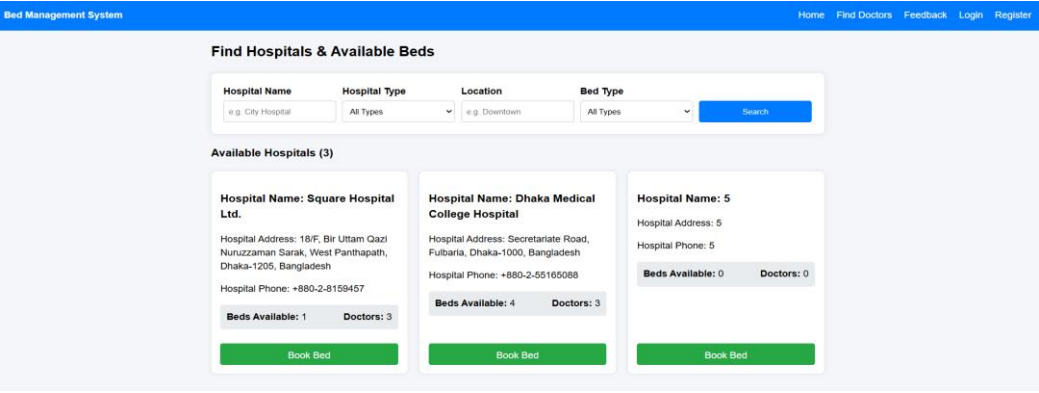
This ER diagram models a Bed Management System connecting hospitals, doctors, beds, patients, admissions, payments, notifications, and feedback. It automates hospital workflows by linking patients to specific doctors and beds during admission. The system seamlessly handles transaction payments, sends automated updates to users, and collects patient feedback to improve healthcare service delivery.

Use Case Diagram



This Use Case Diagram outlines user interactions within the Bed Management System for two primary actors: Hospital and Patient. Hospitals manage administrative tasks like bed allocation, doctor details, and admissions processing. Patients utilize the platform to handle registration, view doctor details, request admissions, complete payments, track notifications, and submit feedback.

UI (Figma/Sketch)



Bed Management System

HomeRegister

Login

Login As: ☒ Patient ☐ Hospital

Patient Username

1

Password

•

Login

Don't have an account? [Register Here](#)

Bed Management System

HomeLogin

Create Account

Account Type: ☒ Patient ☐ Hospital

Username

Unique username

Full Name

John Doe

Phone Number

+1-555-0000

Address

Residential address

Password

Account password

Register

Already registered? [Sign In](#)

Bed Management System

HomeNotificationsLogout

Hospital Management Dashboard

Facility Settings & Password

User Hospital: 5 | Phone: 5 | Location: 5

0

Total Registered Beds

0

Available Beds

0

Medical Staff (Doctors)

0

Pending Admissions

Manage Doctors

Manage Beds

Process Admissions

Edit Profile

4 | Page

Bed Management System

HomeDashboardLogout

Manage Profile & Account Settings

Update Profile Details

Hospital Name

Phone Number

Address

Hospital Details

Update Details

Reset Password

Current Password

Enter current password

New Password

Enter new password

Reset Password

Delete Account

Warning: Deleting your account will permanently wipe your registration and profile record from the system.

Delete Account

Bed Management System

HomeDashboardLogout

Hospital Doctors Directory

Back to Dashboard

Add New Doctor

Doctor Full Name

Dr. Jane Smith

Specialty

e.g. Cardiology

Phone Number

+1-555-0199

Add Doctor

Registered Doctors

Doctor Name	Specialty	Phone	Action
No doctors registered yet.			

Hospital Dashboard - Manage Beds

DashboardLogout

Manage Hospital Beds

Add New Bed

Bed Number (e.g. B-101)

General

Price per day

Add Bed

Existing Beds

Bed #	Type	Price (Tk)	Status	Action
No beds registered yet.				

5 | Page

Hospital Dashboard

DashboardLogout

Process Patient Admissions

ID	Patient Name	Doctor	Bed #	Admission Date	Amount (Tk)	Status	Payment Details	Actions
No pending or existing admission requests found.								

Bed Management System

HomeFind DoctorsNotificationsLogout

Patient Control Center

Patient Profile: 1

Edit Details / Reset Password

Username: 1Phone: 1Address: 1

My Admission History & Status

+ New Admission Request

Admission #	Hospital	Doctor	Bed Info	Date	Amount	Status	Payment Details	Action
#A3	Square Hospital Ltd.	1	Bed #c-1 (Cabin)	2026-09-01	Tk1,200.00	Approved	Payment Method: Mobile Banking Provider: bkash Mobile Number: 01835064304 Transaction ID: jhhbjhb54654	Paid
#A2	Square Hospital Ltd.	2	Bed #g-1 (General)	2026-09-01	Tk500.00	Approved	Payment Method: Mobile Banking Provider: Nagad Mobile Number: 01835064304 Transaction ID: jhhbjhb54654	Paid
#A1	Square Hospital Ltd.	1	Bed #c-1 (CCU)	2026-09-01	Tk2,000.00	Approved	Payment Method: Cash on Place Notice: You will receive a call from the hospital in 5 minutes.	Paid

Bed Management System

HomeDashboardLogout

Manage Profile & Account Settings

Update Profile Details

Full Name

1

Phone Number

1

Address

1

Update Details

Reset Password

Current Password

Enter current password

New Password

Enter new password

Reset Password

Delete Account

Warning: Deleting your account will permanently erase your registration and profile record from the system.

Delete Account

Book Hospital Bed & Admission

1. Select Hospital

Dhaka Medical College Hospital

Dhaka Medical College Hospital

Type: Public

Address: Dharmapala Road, Paltan, Dhaka-1000, Bangladesh

Phone: +880-2-55153581

Details: Tertiary level public teaching hospital offering emergency, surgery, cardiology, pediatrics, and general medicine services.

Hospital Location



2. Select Attending Doctor

Dr. Tanvir

3. Select Available Bed

Bed #1 (CCU)

4. Admission Date

2026-09-04

5. Medical Details / Reason

Covid-19

Save & Proceed to Payment

Payment Checkout

Complete bed booking transaction for Admission #A4

Hospital: Dhaka Medical College Hospital

Doctor: 1

Bed Info: Bed #1 (CCU)

Admission Date: 2026-09-04

Total Payable: Tk1,200.00

Select Payment Method:

Mobile Banking

Mobile Banking Details

Provider: bKash

Mobile Number: 01835064304

Transaction ID: jhbjb54654

Reminder: Ideal: Closing Confirm & Pay complete the payment situation.

Show Details

Full Details:

Payment Method: Mobile Banking
Provider: bKash
Mobile Number: 01835064304
Transaction ID: jhbjb54654

Confirm & Pay Tk1,200.00

Patient Control Center

Patient Profile: 1

[Edit Details / Reset Password](#)

Username: 1 Phone: 1 Address: 1

My Admission History & Status

[+ New Admission Request](#)

Admission #	Hospital	Doctor	Bed Info	Date	Amount	Status	Payment Details	Action
#A4	Dhaka Medical College Hospital	1	Bed #1 (CCU)	2026-09-04	Tk1,200.00	Pending	Payment Method: Mobile Banking Provider: bKash Mobile Number: 01835064304 Transaction ID: jhbjb54654	Paid Cancel
#A3	Square Hospital Ltd.	1	Bed #1 (Cabin)	2026-09-01	Tk1,200.00	Approved	Payment Method: Mobile Banking Provider: bKash Mobile Number: 01835064304 Transaction ID: jhbjb54654	Paid
#A2	Square Hospital Ltd.	2	Bed #1 (General)	2026-09-01	Tk500.00	Approved	Payment Method: Mobile Banking Provider: bKash Mobile Number: 01835064304 Transaction ID: jhbjb54654	Paid
#A1	Square Hospital Ltd.	1	Bed #1 (CCU)	2026-09-01	Tk2,000.00	Approved	Payment Method: Cash on Place Notice: You will receive a call from the hospital in 5 minutes.	Paid

System Notifications

[Mark All as Read](#)

-  Your request for Bed #c-1 (Admission #A4) has been submitted successfully.
 2026-09-04 22:00:48
-  Your request for Bed #c-1 (Admission #A3) has been submitted successfully.
 2026-09-01 09:38:09
-  Your request for Bed #g-1 (Admission #A2) has been submitted successfully.
 2026-09-01 08:58:09
-  Your request for Bed #c-1 (Admission #A1) has been submitted successfully.
 2026-09-01 08:29:50

Hospital Management Dashboard

User Hospital: **Dhaka Medical College Hospital** | Phone: +880-2-55165088 | Location: Secretariate Road, Fulbaria, Dhaka-1000, Bangladesh

[Facility Settings & Password](#)

4

Total Registered Beds

4

Available Beds

3

Medical Staff (Doctors)

1

Pending Admissions

[Manage Doctors](#)

[Manage Beds](#)

[Process Admissions](#)

[Edit Profile](#)

Process Patient Admissions

ID	Patient Name	Doctor	Bed #	Admission Date	Amount (Tk)	Status	Payment Details	Actions
#A4	1	1	Bed #c-1	2026-09-04	Tk1,200.00	Pending	Payment Method: Mobile Banking Provider: bKash Mobile Number: 01835064304 Transaction ID: jhhgh54654	Approve Reject
							Paid	

Process Patient Admissions

Admission approved successfully.

ID	Patient Name	Doctor	Bed #	Admission Date	Amount (Tk)	Status	Payment Details	Actions
#A4	1	1	Bed #c-1	2026-09-04	Tk1,200.00	Approved	Payment Method: Mobile Banking Provider: bKash Mobile Number: 01835064304 Transaction ID: jhhgh54654	Completed
							Paid	

Patient Control Center

Patient Profile: 1

Edit Details / Reset Password

Username: 1 Phone: 1 Address: 1

My Admission History & Status

+ New Admission Request

Admission #	Hospital	Doctor	Bed Info	Date	Amount	Status	Payment Details	Action
#A4	Dhaka Medical College Hospital	1	Bed #c-1 (CCU)	2026-09-04	Tk1,200.00	Approved	Payment Method: Mobile Banking Provider: bKash Mobile Number: 01835064304 Transaction ID: jhghb54654	Paid
#A3	Square Hospital Ltd.	1	Bed #c-1 (Cabin)	2026-09-01	Tk1,200.00	Approved	Payment Method: Mobile Banking Provider: bKash Mobile Number: 01835064304 Transaction ID: jhghb54654	Paid
#A2	Square Hospital Ltd.	2	Bed #g-1 (General)	2026-09-01	Tk500.00	Approved	Payment Method: Mobile Banking Provider: Nagad Mobile Number: 01835064304 Transaction ID: jhghb54654	Paid
#A1	Square Hospital Ltd.	1	Bed #c-1 (CCU)	2026-09-01	Tk2,000.00	Approved	Payment Method: Cash on Place Notice: You will receive a call from the hospital in 5 minutes.	Paid

How are you feeling?

Your feedback helps us improve our services.

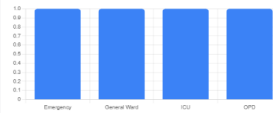


Hospital Name:
 Department:
 Comment:
☐ Remember my name (Cookie)

Submit Feedback

Home

Department Analytics



Recent Feedbacks

#	Name	Dept	Details
#4	Bedanta saha	OPD	[Rating: 3/5] bad
#3	Bedanta saha	General Ward	[Rating: 3/5] good
#2	Bedanta saha	ICU	[Rating: 3/5] good
#1	Bedanta saha	Emergency	[Rating: 3/5] no comment



> controllers

> Database

> models

> views

index.php